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Course: Data Science

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Exercise: 2

Topic: SQL Aggregate Functions & Operators

1. SELECT DISTINCT department
FROM students;

department
IT
HR
Finance

2. SELECT department, AVG(age) AS avg_age
FROM students
GROUP BY department;

Calculation : IT = (20 + 21)/2 = 20,5
HR = (22 + 22)/2 = 22
Finance = (23 + 0)/1 = 23

department	avg_age
IT	20,5
HR	22
Finance	23

3. SELECT department, COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;

department	student_count
IT	2
HR	2

```
4. SELECT student_id, name, age, department
   FROM   students
   WHERE  age BETWEEN 21 AND 23;
```

student_id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
4	Diana	23	Finance
5	Eve	22	HR

```
5. SELECT student_id, name, age, department
   FROM   students
   WHERE  department IN ('IT', 'HR')
   AND    age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

```
6. SELECT department, COUNT (*) AS total_credits
   FROM   courses
   GROUP BY department
   HAVING COUNT (*)
```

```
6. SELECT department, SUM (credits) AS total_credits
   FROM   courses
   GROUP BY department
   HAVING SUM (credits) > 5;
```

department	total_credits
IT	11

7. SELECT *
FROM courses
WHERE credits < 4;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id, course_name, credits
FROM ~~credit~~ courses
GROUP BY credits DESC,
LIMIT 3;

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. SELECT MAX (grade) AS max_grade,
MIN (grade) AS min_grade,
AVG (grade) AS avg_grade
FROM enrollments;

Calculation for AVG (grade) $= \frac{85 + 78 + 90 + 88 + 82}{5}$

$= 84,6$

max_grade	min_grade	avg_grade
90	78	84,6

10.
```
SELECT   course_id , COUNT (*) AS total_enrollments
FROM     enrollments
GROUP BY course_id ;
```

course_id	total_enrollments
101	1
102	1
103	1
104	1
105	1

11.
```
SELECT   department,
         SUM (salary) AS total_salary,
         SUM (bonus) AS total_bonus
FROM     salaries
GROUP BY department ;
```

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6000

12.
```
SELECT   department , AVG (salary) AS avg_salary
FROM     salaries
GROUP BY department
HAVING   AVG (salary) > 55 000 ;
```

department	avg_salary
IT	61 000
Finance	70 000

13.
```
SELECT employee_id, name, salary, bonus, SUM (salary) + SUM (bonus)
        AS total_compensation
FROM salaries
GROUP BY name
HAVING [SUM (salary) + SUM (bonus)] > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

14.
```
SELECT department, SUM (budget) AS total_budget,
        AVG (budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG (budget) > 70 000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. ~~SELECT project_id, project_names, departments, budget~~

15.
```
SELECT *
FROM projects
WHERE NOT department = 'Marketing'
AND buget BETWEEN 50 000 AND 120 000;
```

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
5	HR Portal	HR	50 000